

Complexity of Deep Computations via Topology of Function Spaces

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What happens at infinite depth?

Neural networks are getting **deeper** year by year:

AlexNet (2012): 8 layers $\longrightarrow$ ResNet (2015): 152 layers $\longrightarrow$ modern LLMs: >100 transformer blocks.

Thought experiment 1. **Depth** $\rightarrow \infty$ seen as **continuous-time** $t \rightarrow \infty$:

- **Neural ODEs** Chen *et al.* [Che+18]: Replace **discrete layers** with a **continuous-time ODE** for the state-evolution function $\mathbf{z}(t) = \mathbf{z}_\theta(t) = \mathbf{z}(\theta, \mathbf{z}_0; t)$ defined by the **autonomous** ODE:

$$\frac{d}{dt} \mathbf{z}(t) = \mathbf{f}_\theta(\mathbf{z}(t)), \quad \mathbf{z}(0) = \mathbf{z}_0, \quad t \in [0, \infty), \quad \theta: \text{real parameter(s)}.$$

- “**Deep Equilibrium states**” as $t \rightarrow \infty$ become *asymptotic equilibria* of trajectories.
 - *Deep Equilibria* for parameter θ are any states $\mathbf{z}^* = \mathbf{z}^*(\theta)$ **implicitly** defined by:

$$\mathbf{f}_\theta(\mathbf{z}^*) = \mathbf{0}$$

- Found (in practice) using a “black-box” *iterative root finder* **RootFind** for $\mathbf{f}_\theta$ with **initial condition** $\mathbf{z}_0$.

Use of a RootFind algorithm “**short-circuits**”/circumvents forward-propagation through the network!

What happens at infinite depth? (Continued...)

Thought experiment 2. **Depth** $\rightarrow \infty$ as a **fixed-point problem** under (**discrete-time**) iterations:

- **Deep Equilibrium Models (DEQs)** [BKK]: Iterate “**weight-tied**” layers (successive layers with identical weights) *ad infinitum*.
- If the layer transformation is f_θ :

$$z_{n+1} = f_\theta(z_n), \quad \text{given } \theta, z_0: \text{ real parameters,}$$

until reaching a **fixed point** $z^* = z^*(\theta)$ implicitly defined by

$$z^* = f_\theta(z^*).$$

- A black-box fixed-point algorithm **FixFind** for f_θ seeded with given **initial** z_0 iteratively finds some **equilibrium** $z^*(\theta)$.

Central questions of this talk

What **kind of functions** $\theta \mapsto z^*(\theta; z_0)$ does an infinitely deep network compute? When is $z^*(\theta; z_0)$ **effectively computable**? When is $z^*(\theta; z_0)$ a **tame** vs. **wild** function?

Deep Equilibrium Models: A worked example

Fix the **activation function** $f_{\theta}(z) = \tanh(\mathbf{w}z + \mathbf{b})$.

We follow the *implicit/ugly* convention that θ stands for “all adjustable parameters”; effectively, $\theta = (\mathbf{w}, \mathbf{b})$ here.

Forward pass: Iterate until convergence

$$z_{n+1} = \tanh(\mathbf{w} z_n + \mathbf{b}) \quad \text{starting from given } z_0.$$

- 1, 2 or 3 **fixed points** z^* exist if $w \neq 0$.
- Fixed points may be found using a “**black box**” **iterative algorithm** different from the recursive definition of z_n .
- **Which** fixed point z^* is found starting from z_0 **depends on the algorithm**.

Backward pass (implicit diff’n):

$$\frac{\partial z^*}{\partial \mathbf{w}} = \frac{(1 - z^{*2}) z^*}{1 - \mathbf{w}(1 - z^{*2})}, \quad \text{etc.}$$

No “**infinite**” **back-propagation** needed.

- Why this matters:
 - **$O(1)$ memory**, regardless of “effective depth”;
 - **Gradient computation is *exact* at the limit**;
 - **Bai–Kolter–Koltun (2019)**: Empirically stable.

BKK leave open: Existence of z^* and what kind of function $\theta \mapsto z^*(\theta; z_0)$ is.

Infinite-depth limits are Baire class 1 functions

Standard (finite-depth) networks compute **continuous** functions.

Infinite-depth limits arise from sequences of continuous functions as **pointwise limits** or **pointwise ultralimits**: distinguishing limits vs. ultralimits is **critical**.

Definition (Baire class 1)

A **real function** g on a topological space X is called **Baire class 1** (Baire-1, $B_1(X)$ or simply B_1) if it is the **pointwise limit** $g = \lim_{n \rightarrow \infty} f_n$ of a sequence of *continuous functions* f_n on X .

Baire-1

- “Tame” deep equilibria;
- Thomae’s (ruler) function;
- Newton’s root-finding algorithm
 $(f; z_0) \mapsto \text{Newton}(f; z_0)$ for $z_0 \in \text{Fatou}$.

Not Baire-1

- “Wild” deep equilibria;
- Dirichlet’s function $1_{\mathbb{Q}}$;
- Julia-set indicator functions, e.g.,
 $(f; z_0) \mapsto \text{Newton}(f; z_0)$ for $z_0 \in \text{Julia}$.

Main questions restated: Which deep computations are **genuine pointwise limits** in B_1 ?
Are there **computational complexity sub-subclasses** within B_1 ?

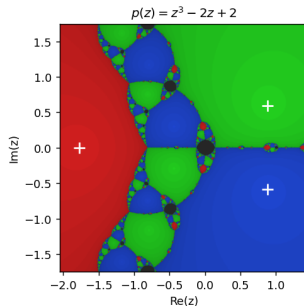
A running example: Newton's method as deep computation

Newton's method: Solve $p(z) = 0$ for a given **polynomial** $p = p_\theta$ by successive applications of the *Newton map* $N_\theta = N_p$:

$$z_{n+1} = N_p(z_n), \quad z_0 \in \mathbb{C} \text{ given}; \quad N_p(z) := z - \frac{p(z)}{p'(z)}.$$

The extended complex plane $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ splits into:

- **Tame** dynamics: **Fatou set**.
 - Dense open set of initial points z_0 having a neighborhood where a sub-sequence of Newton iterates converges pointwise.
- **Wild** dynamics: **Julia set** (Fatou's complement).
 - Points z_0 having *no* neighborhood whereon a subsequence of iterates converges pointwise.

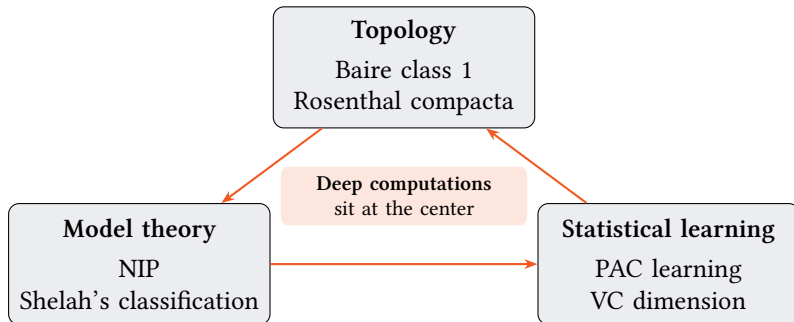


Newton fractal for $z^3 - 2z + 2$.

Any **ultralimit** $z^*(z_0; \theta) = z_\omega^*(z_0; \theta) = \mathcal{U}\lim_{n \rightarrow \infty} z_n$ is a **deep computation** or **ultra-computation**.

The Rosetta Stone: Three classification programs

The same tameness/wildness dichotomy appears **independently** in three fields:



Tame deep computations are simultaneously:

Baire-1 (topology) $\iff$ NIP (model theory) $\iff$ PAC-learnable (statistics)

Roadmap of the talk

§2 A mathematical framework for deep computation

Computation States Structures (CSSs) and Compositional Computation Structures (CCSs); the type map $\text{tp}: L \rightarrow \mathbb{R}^{\mathbb{N}}$.

§3 Existence of deep equilibria

Deep computations as ultrafilter limits; the Ellis–Numakura Lemma guarantees existence for confined computations.

§4 The tame/wild dichotomy: Baire class and NIP

When is a deep computation in B_1 ? The Bourgain–Fremlin–Talagrand theorem as a computable criterion.

§5 A finer classification: levels of tameness

Todorćević’s trichotomy for Rosenthal compacta; NIP levels and PAC-learnability; Newton fractals as a running example.

§6 Conclusion and open problems

A model for computation at infinite depth

Fix a **state space** L and a countable family $\mathcal{P} = (P_n)$ of **real-valued observables** $P_n: L \rightarrow \mathbb{R}$ (coordinates, norms, layer outputs, ...).

The **type** of $v \in L$ encodes all its **observable features**:

$$\text{tp}(v) := (P_n(v))_{n \in \mathbb{N}} \in \mathbb{R}^{\mathbb{N}}.$$

The **type space** $\bar{L} = \overline{\text{tp}(L)} \subseteq \mathbb{R}^{\mathbb{N}}$ (closure in the product topology) adjoins “**ideal**” **ultrastates** reachable only at *infinite depth*.

A **computation** $\gamma: L \rightarrow L$ is any **continuous** state transformation; computations compose, forming a semigroup Γ .

CSS and CCS (in brief)

- **CSS** = $\langle L, \mathcal{P} \rangle$: state space + observables.
- **CCS** = $\langle L, \Gamma \rangle$: CSS + semigroup of computations.
- When L is **compact**, so is $\bar{L}$ (by Tychonoff): **no escape to infinity**.

Running examples

- **DEQ**. $L = \mathbb{R}$; $\gamma_{\theta}(z) = \tanh(wz + b)$.
- **Newton**. $L = \hat{\mathbb{C}}$ (extended complex plane, **compact**); $\gamma = N_p$.

Deep computations: ultrafilter limits of iterations

Iterate a single computation indefinitely: $\gamma^0 = \text{id}$, $\gamma^1 = \gamma$, $\gamma^2 = \gamma \circ \gamma$, ...

A **deep computation** (ultracomputation) is any **ultrafilter limit**

$$\gamma^{(\mathcal{U})} := \mathcal{U}\lim_{n \rightarrow \infty} \gamma^n,$$

defined **pointwise**: $\gamma^{(\mathcal{U})}(\nu) = \mathcal{U}\lim_n \gamma^n(\nu) \in \bar{L}$.

- $\gamma^{(\mathcal{U})}(\nu)$ may be an **ultrastate**, i.e., in $\bar{L} \setminus L$.
- $\gamma^{(\mathcal{U})}$ is **typically discontinuous**.
- When $L = \bar{L}$ is **compact**: every ultrafilter $\mathcal{U}$ on $\mathbb{N}$ gives a “realized” deep computation $\gamma^{(\mathcal{U})} : L \rightarrow L$.

Finite computations are **continuous** (Baire class 0). Deep computations belong to **Baire class 1 or higher**.

Baire class 1 (B_1)

A function $g: L \rightarrow \mathbb{R}$ is **Baire-1** if

$$g = \lim_{n \rightarrow \infty} f_n$$

pointwise, with each f_n **continuous**.

Baire-1 means g is “barely” **discontinuous**.

Central questions for this talk:

- When is a **deep computation** **Baire-1**?
- What sequences of computations yield **Baire-1** deep computations —but **no higher**?

Deep equilibria exist — Ellis–Numakura

Definition (Deep equilibrium)

A deep computation $f_{\mathcal{U}}$ of γ is a **deep (iterative) equilibrium** if it is **idempotent**:

$$f_{\mathcal{U}} \circ f_{\mathcal{U}} = f_{\mathcal{U}}.$$

Theorem ([Alv+24], via Ellis–Numakura [Ell58])

*If the orbit closure $\overline{\{\gamma^n : n \geq 1\}}$ is **compact**, at least one deep equilibrium exists.*

▷ A compact topological semigroup always contains an **idempotent** element —this is the Ellis–Numakura Lemma (Zorn’s-lemma argument on minimal closed sub-semigroups). $\square$

Newton: $L = \widehat{\mathbb{C}}$ (compact)

Orbit closure $\overline{(N_p^n)_n}$ compact
 $\Rightarrow$ **DEQ always exist.**

B₁ on Fatou: $\forall \mathcal{U}$ there exists a subsequence (n_i) s.t. $N_p^{(n_i)} = \lim_i N_p^{n_i}$ (on the Fatou set).

DEQ: Orbit stays in $[-1, 1]$

$|f_{\theta}(z)| = |\tanh(\mathbf{w}z + \mathbf{b})| \leq 1 \Rightarrow$ orbit closure $\overline{(f_{\theta}^n)_n}$ **compact** $\Rightarrow$ **DEQ exist.**

Tame dynamics: $\forall \mathcal{U}: (\theta; z) \mapsto f_{\theta}^{(\mathcal{U})}(z)$ is B_1 .

Bourgain–Fremlin–Talagrand: A fundamental dichotomy

Question. Given an iteration (γ^n) , which deep iterates $\gamma^{(\mathcal{U})}$ are **Baire-1**?

Theorem (BFT dichotomy [BFT78])

For a *locally uniformly bounded* family $(\gamma^n)_{n \in \mathbb{N}}$ of continuous functions on a Polish space, *exactly one* of the following holds:

- (Tame)** The closure $\overline{\{\gamma^n\}}$ in the pointwise topology is included in B_1 : *every deep iterate* $\gamma^{(\mathcal{U})} = \lim_i \gamma^{n_i}$ *is a sub-sequential limit.*
- (Wild)** The closure $\overline{\{\gamma^n\}}$ has a subset *homeomorphic to* $\beta\mathbb{N}$ (the Stone–Čech compactification of $\mathbb{N}$).

- **Tame.** There are at most $2^{\aleph_0}$ distinct deep iterates: Any $\mathcal{U}$ gives an ultralimit $\gamma^{(\mathcal{U})} = \lim_i \gamma^{n_i}$ for some subsequence (n_i) .
- **Wild.** Deep iterates form an enormous set of cardinality $2^{2^{\aleph_0}}$: Distinct ultrafilters give distinct deep computations **not arising as sequential limits** of any subsequence.

Independence patterns: the wild side

What distinguishes **tame** from **wild**? An **independence pattern** in (γ^n) .

Definition (NIP, informal)

(γ^n) has the **Independence Property (IP)** at threshold $a < b$ if for **every finite disjoint** $E, F \subseteq \mathbb{N}$ there exists a state v with

$$\gamma^n(v) \leq a \text{ for } n \in E, \quad \gamma^n(v) \geq b \text{ for } n \in F.$$

(γ^n) is **NIP** (“**No IP**”) otherwise.

Note that **NIP** is the “**good**” property that *bounds* the complexity of (γ^n) .

For a family $\mathcal{F}$ of $\{0, 1\}$ -valued functions:

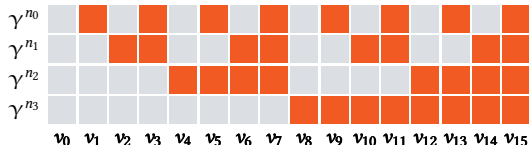
$\mathcal{F}$ has **IP** $\Leftrightarrow \text{VC-dimension}(\mathcal{F}) = \infty$

$\Leftrightarrow \mathcal{F}$ **shatters** arbitrarily large finite sets.

Picture of IP

A (hypothetical) 4-row independence pattern for **all partitions**

$$E \sqcup F = \{n_1, n_2, n_3, n_4\}.$$



IP family $\mathcal{F} = \{\gamma^{n_0}, \gamma^{n_1}, \gamma^{n_2}, \gamma^{n_3}\}$: **All** $2^4 = 16$ sign-patterns

Tame families $\mathcal{F}$: Fewer than $2^{\#\mathcal{F}}$ patterns.

Three equivalent faces of tameness (Main Theorem)

Theorem (Cornerstone of the paper)

Let $(L, \mathcal{P}, \Gamma)$ be a CCS with $\mathcal{P}$ countable and $\Delta \subseteq \Gamma$ a **confined family** of continuous computations. Then the following are equivalent:

- ❶ **(Topology)** $\overline{\Delta}$ is a **Rosenthal compactum** (i.e., $\overline{\Delta} \subseteq B_1$);
- ❷ **(Model Theory)** For each observable $P \in \mathcal{P}$, the family $(P \circ \gamma)_{\gamma \in \Delta}$ is **NIP**;
- ❸ **(Statistical Learning)** The $\{0, 1\}$ -valued “**level-functions**” of Δ have **finite VC dimension**, hence are **PAC-learnable**.

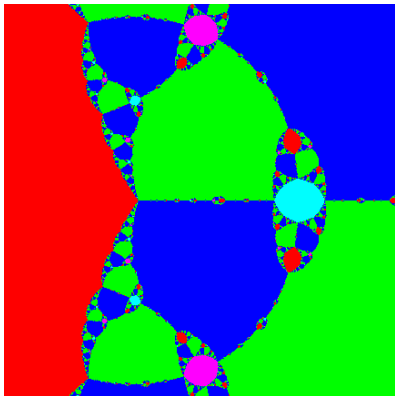
Moreover, in that case **every** deep iterate $\gamma^{(u)} \in \overline{\Delta}$ is a **sequential pointwise limit** of computations in Δ .

$$\text{Tame} = \text{Baire-1} = \text{NIP} = \text{PAC-learnable}$$

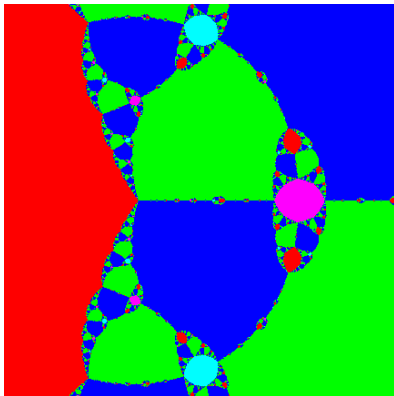
Slogan. “Deep computation is feasible **iff** the underlying computation family is statistically learnable.”

Newton fractals: tame, but with structure

Recall. $p(z) = z^3 - 2z + 2$ has 3 roots r_0, r_+, r_- and 2-cycle $0 \leftrightarrow 1$ of N_p (not roots of p).



$n = 100$ (even)



$n = 101$ (odd)

R, G, B regions: basins of the three actual roots $r_0, r_{\pm}$.

Cyan / magenta regions: Basin of the 2-cycle $\{0, 1\}$.

Cyan $\rightleftharpoons$ magenta when $n \rightarrow n + 1$: Even/odd subsequences $(N_p^{2i}), (N_p^{2i+1})$ converge to **distinct** deep equilibria f_{ev} and f_{odd} .

The tame world is still richer than “one limit”

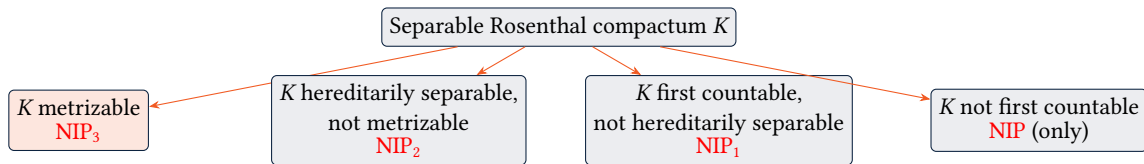
Both deep equilibria $f_{\text{ev}}, f_{\text{odd}}$ are **Baire-1**; together, they capture **asymptotic dynamical structure**.

Inside Baire-1: Todorčević's trichotomy

Even *within* the tame class, Baire-1 has a **fine internal structure**.

Theorem (Todorčević's trichotomy [Tod99])

*Every **separable** Rosenthal compactum K that is non-metrizable contains a canonical copy of **one of three minimal prototypes**.*



- Strict implications: $\text{NIP}_3 \Rightarrow \text{NIP}_2 \Rightarrow \text{NIP}_1 \Rightarrow \text{NIP}$.
- For deep computations: four levels of **topological tameness** refining “Baire-1”.
- Further refinement: Argyros–Dodos–Kanellopoulos **heptachotomy**: **seven** prototypes.

Examples at each level of tameness

Each NIP_i level is non-empty: There are concrete CCSs realizing each.

Level	CCS	Closure of Δ in B_1
NIP_3	Square-root iteration (compute $\sqrt{x}$)	Alexandroff compactification (metrizable)
NIP_2	Finite-precision threshold classifiers on $2^{\mathbb{N}}$	Split Cantor $S(2^{\mathbb{N}})$ (her. sep. but not metrizable)
NIP_1	Bit-shift family $\{\gamma_t : t \in 2^{<\mathbb{N}}\}$	$\widehat{D}(S(2^{\mathbb{N}}))$ (1st countable, not her. sep.)
NIP (only)	Finite-precision prefix tests	Extended Alexandroff $\widehat{A}(2^{\mathbb{N}})$

Interpretation. Each level corresponds to **how efficiently** deep computations can be **approximated** by standard ones:

- NIP_3 (metrizable): countable, **near-uniformly** convergent;
- $\text{NIP}_2 \rightarrow \text{NIP}$: progressively **harder** approximation.

Beyond NIP: The wild side

IP families: Closure contains $\beta\mathbb{N}$ (consists of $2^{2^{\aleph_0}}$ **distinct** deep computations).

No reduction to a countable Δ : **Not approximable** by sequences.

Summary

Framework (1st paper: Alva–Dueñez–Iovino–Walton 2024)

- **CSSs** and **CCSs**: A topological framework for real-valued computation.
- **Deep computations** arise as ultrafilter limits in state type space.
- **Existence of deep equilibria** follows from compactness + Ellis–Numakura.

Classification (2nd paper: Dueñez–Iovino–Matos–Salvetti–Tall)

- **BFT dichotomy**: Tame (Baire-1) vs. Wild (contains $\beta\mathbb{N}$).
- **Three-fold equivalence**: **Baire-1** $\Leftrightarrow$ **NIP** $\Leftrightarrow$ **finite VC** ($\Leftrightarrow$ PAC-learnable).
- **Baire-1 sub-classes**: Todorčević's trichotomy ($\text{NIP}_3 \Rightarrow \text{NIP}_2 \Rightarrow \text{NIP}_1 \Rightarrow \text{NIP}$ hierarchy). (Argyros–Dodos–Kanellopoulos: 7 minimal prototypes).

Deep computation is feasible $\Leftrightarrow$ the underlying family is statistically learnable.

Open problems and directions

- **Which neural architectures are NIP?** Identify NIP/IP for concrete DEQ and Neural ODE families; characterize “computationally feasible” deep architectures.
- **Monte Carlo deep computability.** Probabilistic versions of NIP (**Talagrand stability**); Monte Carlo computation of deep equilibria *when* pointwise convergence fails.
- **Non-topological characterizations of NIP_i .** Can the $\text{NIP}_1, \text{NIP}_2, \text{NIP}_3$ levels be detected by purely **combinatorial** or **logical** invariants of Δ ?
- **In-context learning in transformers.** Modern LLMs exhibit *learning at inference time*. Are the in-context hypothesis classes NIP? Finite VC?
- **Beyond NIP.** Refinements of IP (NTP_2 , simplicity, distality) yield a finer wild taxonomy. Are there natural hierarchies of **wild** deep computations?

¡Muchas gracias!

Questions and comments welcome.

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